LAB Assignment-4

31-08-2025

m = 3

machines = A, B, C

No of Plays = 10

Machine	Rewards
A	3 4 2 5
B	6 7 5 8
C	4 3 5 4

$$UCB(a) = Q(a) + \sqrt{\frac{2\ln(t)}{N(a)}}$$

$$Q_{new}(a) = Q_{old}(a) + \frac{R - Q_{old}(a)}{N(a)}$$

$$R_{total} = \sum R$$

Initialization

machine	N	R	Q
A	0	0	0
B	0	0	0
C	0	0	0

play = 1   R = 3   N(A) = 1

New table

Machin	N	R	Q
A	1	3	3
B	0	0	0
C	0	0	0

Play 2 - Machine B

$R = 6$

$N(B) = 1$

$Q(B) = 0 + \frac{6-0}{1}$

$\boxed{Q(B) = 6}$

Play 3 = Machine C

$R = 4$

$N(C) = 1$

$Q(C) = 0 + \frac{4-0}{1}$

$Q(C) = 4$

Machine	N	R	Q
A	1	3	3
B	1	6	6
C	1	4	4

Play 4 =

$$UBC(A) = Q(A) + \sqrt{\frac{2\ln(4)}{n_A}}$$

$$3 + \sqrt{\frac{\ln(6)}{1}} = 5.35$$

$$UBC(B) = 6 + 2\sqrt{\frac{\ln(6)}{1}} = 8.35$$

$$UBC(C) = 6 + 2\left\{\sqrt{\frac{\ln(4)}{1}}\right\} = 6.65$$

B ✓

Update N = $1+1=2$

Total Reward = $6+7=13$

Update Q = ~~$6 + \frac{R - Q}{n}$~~ $6 + \frac{7-6}{2}$

$6 + \frac{1}{2}$

$\boxed{6.5}$

Current Value

	N	R	Q
A	1	3	3
B	2	13	6.5
C	1	4	4

UCB (A)

$$UCB(A) = 3 + \sqrt{\frac{2\ln(5)}{1}}$$

$$3 + \sqrt{3.2189}$$

$$= 3 + 1.794$$

$$UCB(A) = 4.794$$

UCB(B) = 7.769 B is selected.

UCB(C) = 5.794

$R = 5$

$N(B) = 2+1 = 3$

$R(B) = 13+5 = 18$

$$6.5 + \frac{5 - 6.5}{3}$$

$$6.5 + \frac{-1.5}{3}$$

$6.5 - 0.5$

$\boxed{Q(B) = 6}$

Play 6

A

$UCB(A) = 3 + \sqrt{2 \ln 6}$

$= 3 + \sqrt{3.58 35}$

$\boxed{UCB(A) = 4.896}$

B

$UCB(B) = 7.093$

C

$UCB(C) = 5.893$

$R = 8$

$N(B) = 4$

$R(B) = 18+8 = 26$

$$Q(B) = 6 + \frac{8-6}{4}$$

$6 + 2/4$

$Q(B) = 6.5$

Play = 7

$N_B = 7$

$Q_B = 6.5$

$UCB(B) = 6.5 + \sqrt{\frac{2\ln(7)}{4}}$

$6.5 + \sqrt{\frac{3.8918}{4}}$

$6.5 + \sqrt{0.97295}$

$6.5 + 0.986$

$\boxed{UCB(B) = 7.486}$

B selected

R = 6

N(B) = 5

R(B) = 26 + 6 = 32

$Q(B) = 6.5 + \frac{6 - 6.5}{5}$

$6.5 - \frac{0.5}{5}$

$Q(B) = 6.4$

Play 8 =

t = 8

$UCB = 6.4 + \sqrt{\frac{2\ln 8}{5}}$

$6.4 + \sqrt{\frac{4.1589}{5}}$

$6.4 + \sqrt{0.8318}$

$6.4 + 0.912$

$\boxed{UCB(B) = 7.312}$

R = 7

N(B) = 6

R(B) = 32 + 7 = 39

$Q(B) = 6.4 + \frac{7 - 6.4}{6} = 6.4 + \frac{0.6}{6} = Q(B) = 6.5$

Play 9 =

t = 9

$UCB = 6.5 + \sqrt{\frac{2\ln 9}{6}}$

$6.5 + \sqrt{\frac{6.3944}{6}}$

$6.5 + \sqrt{0.7324}$

$6.5 + 0.856$

$UCB(B) = 7.356$

update R = 5

N(B) = 7

R(B) = 39 + 5 = 44

$Q(B) = 6.5 + \frac{5 - 6.5}{7}$

$6.5 - \frac{1.5}{7}$

$Q(B) \approx 6.286$

Play 10 = $t=10$

$UCB(B) = 6.286 + \sqrt{\frac{2\ln(10)}{7}}$

$6.286 + \sqrt{\frac{6.6052}{7}}$

$6.286 + 0.811$

select B

$R = 8$

$N(B) = 8$

$R(B) = 44 + 8 = 52$

$Q(B) = 6.286 + \frac{8 - 6.280}{8}$

$6.286 + \frac{1.714}{8}$

$Q(B) \approx 6.5$

Final Table

A	1	3	3.00
B	8	52	6.50
C	1	4	4.00

t = 10

$UCB(A) = 3 + \sqrt{\frac{2\ln(10)}{1}}$

$= 3 + 2.146$

$UCB(A) = 5.146$

$UCB(B) = 6.5 + \sqrt{\frac{2\ln 10}{8}}$

$6.5 + \sqrt{0.57565}$

$6.5 + 0.759$

$UCB(B) = 7.259$

$UCB(C) = 4 + \sqrt{\frac{2 \ln 10}{1}}$

$4 + 2.146$

$UBC(C) = 6.146$

Machine	N	R	UCB	Q
A	1	3	5.146	3.00
B	8	52	7.259	6.50
C	1	4	6.146	4.00

Rewards

$= 3 + 6 + 9 + 7 + 5 + 8 + 6 + 7 + 5 + 1$

$\boxed{R_{total} = 59}$